INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Pembrolizumab 400 mg + weekly carboplatin AUC 1.5 / paclitaxel → dose-dense AC

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
Intervals differ by treatment phase

Treatment stages
4
Explained in the treatment reference

Regimen source
NCCP 00861
Version 2

Treatment stages

01 **Phase 1 · carboplatin and paclitaxel**
About 12 weeks

02 **Phase 2 · dose-dense AC**
About 8 weeks

03 **Surgery and pathology**
After recovery

04 **After surgery · pembrolizumab alone**
Separate phase

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

Pembrolizumab 400 mg + weekly carboplatin AUC 1.5 / paclitaxel → dose-dense AC
National Cancer Control Programme (NCCP) 00861

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Report new breathlessness, chest symptoms, severe diarrhoea, inability to drink, jaundice, a painful swollen leg or a severe rash promptly. You do not need to identify which medicine caused the symptom.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

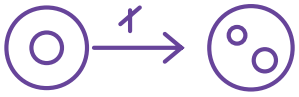
Medicines in the active phase

Treat the cancer before surgery as part of a plan intended to reduce the risk of it returning.

IMMUNOTHERAPY

Pembrolizumab

Keytruda



Releases an immune brake

PD-1 is a control switch on T cells, a type of immune cell. When it connects with PD-L1 or PD-L2, the T cell receives a signal to reduce its activity. Some cancers use this pathway to escape immune attack. Pembrolizumab is an antibody that blocks PD-1, helping T cells recognise and attack the cancer. It does not directly poison cancer cells like chemotherapy. Releasing this brake can also allow immune cells to attack healthy organs, which explains the different side-effect pattern and why symptoms can appear after treatment stops.

Immune cell → Brake blocked → Cancer cell

What to watch for

The immune system can also inflame healthy organs. New cough, diarrhoea, unusual weakness, severe headache, thirst or a spreading rash need prompt advice. Effects can start during treatment or after it finishes.

PLATINUM CHEMOTHERAPY

Carboplatin



Damages deoxyribonucleic acid (the cell's genetic material) (DNA) inside cells

Carboplatin creates links in DNA, making it harder for cells to copy and repair their genetic instructions. Its dose depends on kidney function as well as the intended regimen.

Platinum enters cell → DNA links form → Copying is disrupted

What to watch for

Keep blood-test appointments. Fever, unusual bleeding, repeated vomiting or infusion symptoms need prompt advice.

TAXANE CHEMOTHERAPY

Paclitaxel

Taxol (conventional paclitaxel)



Locks the cell division machinery

Cells use tiny structures called microtubules to separate their chromosomes when dividing. Paclitaxel stabilises these structures so they cannot rearrange normally, interrupting cell division. It can also affect nerves and other healthy tissues.

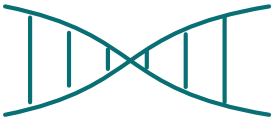
Microtubules form → Structures remain locked → Division is interrupted

What to watch for

Report tingling or numbness before the next weekly dose, especially if it affects walking or hand use. Tell the nurse immediately about flushing, chest or throat tightness, itching, dizziness or breathing changes during infusion.

ANTHRACYCLINE CHEMOTHERAPY

Doxorubicin



Interferes with DNA handling

Doxorubicin interferes with enzymes and DNA processes needed for cell division. It can also affect heart muscle, so the team considers previous exposure and cardiac risk.

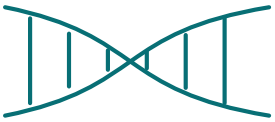
DNA handling disrupted → Cell damage → Division reduced

What to watch for

Report pain or burning at the infusion site immediately. New breathlessness or chest symptoms need prompt review. Red urine for a short time can be a drug-colour effect; pain, clots or persistent bleeding still need advice.

ALKYLATING CHEMOTHERAPY

Cyclophosphamide



Damages DNA needed for growth

Cyclophosphamide is activated in the body and damages DNA. This helps stop dividing cancer cells but can also affect blood-cell production and reproductive tissues.

Drug activated → DNA damage → Cell division affected

What to watch for

Use the prescribed anti-sickness and hydration plan. Report fever, painful urination or blood in the urine.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why might my team recommend this treatment?

This treatment is used for eligible high-risk triple-negative breast cancer before surgery, then continued with immunotherapy after surgery when appropriate. Chemotherapy damages cancer cells, while pembrolizumab helps immune cells recognise and attack them. The team considers cancer extent, fitness, heart health and whether immunotherapy is suitable.

Why more than one phase?

Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks). Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) support is required for every dose-dense AC cycle; the team prescribes its product and timing separately.

What determines the next step?

After surgery: pembrolizumab 400 mg every 42 days for five doses, beginning six weeks after cycle 8 Day 1, subject to recovery, surgery and the oncology plan. Progression preventing definitive surgery or unacceptable pembrolizumab toxicity changes this plan; postoperative treatment is not automatic.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

How the phases connect

Treatment stages

01 **Phase 1 · carboplatin and paclitaxel**
About 12 weeks

02 **Phase 2 · dose-dense AC**
About 8 weeks

03 **Surgery and pathology**
After recovery

04 **After surgery · pembrolizumab alone**
Separate phase

Your team confirms the active stage and dates. The full treatment guide explains each step.

This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

ABOUT 12 WEEKS

Phase 1 · carboplatin and paclitaxel

Four 21-day cycles

Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks).

ABOUT 8 WEEKS

Phase 2 · dose-dense AC

Four 14-day cycles

Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) support is required for every dose-dense AC cycle; the team prescribes its product and timing separately.

AFTER RECOVERY

Surgery and pathology

Individual timing

The operation is planned after recovery; the pathology result helps decide the next treatment.

SEPARATE PHASE

After surgery · pembrolizumab alone

Confirm the prescribed schedule

After surgery: pembrolizumab 400 mg every 42 days for five doses, beginning six weeks after cycle 8 Day 1, subject to recovery, surgery and the oncology plan. Progression preventing definitive surgery or unacceptable pembrolizumab toxicity changes this plan; postoperative treatment is not automatic.

The phases run in order. The columns show sequence, not equal time intervals. No dates or other information need to be entered.

BASELINE → TREATMENT → FOLLOW-UP

When checks happen

Relative sequence, not booked dates or equal time intervals. Baseline electrocardiogram (ECG) is distinct from risk-based echo. No routine three-monthly computed tomography (CT) or echo is implied. The team books and reviews results.

Check	Before starting	Phase 1 · weeks 1–12	Before / during AC	Surgery / postoperative	After treatment
Blood tests	full blood count (FBC), renal and liver; verify GFR	Weekly before planned doses	Before each 14-day AC cycle	Before each pembrolizumab cycle	Symptoms or abnormalities determine follow-up
ECG / echo	ECG and cardiac risk review; echo if indicated	Symptoms prompt earlier assessment	Review before first doxorubicin; further echo by risk / symptoms	Review any cardiac symptoms or previous abnormalities	Risk-based cardiac follow-up
Cancer imaging	Breast/nodal imaging; staging as indicated	Clinical response review; image for concern	Assess response and plan surgery	Pathology review; agree next steps	Breast surveillance; no routine CT/ markers when asymptomatic
Thyroid / hormones	Thyroid baseline; hepatitis screening per source	Thyroid every 3–6 weeks	Continue immune monitoring	Continue while on pembrolizumab	Delayed immune symptoms still require review
Surgery / radiation	Agree overall multidisciplinary team (MDT) pathway	Systemic treatment first	Complete neoadjuvant phase if appropriate	Surgery → pathology → radiation if indicated	Follow-up and late-effect support

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00861 v2 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · ASCO · neoadjuvant breast therapy and response assessment

THE WHOLE TREATMENT JOURNEY

Pembrolizumab 400 mg + weekly carboplatin AUC 1.5 / paclitaxel → dose-dense AC: the whole pathway

Triple-negative breast cancer · neoadjuvant and adjuvant immunotherapy

Phase 1: four 21-day cycles; paclitaxel on Days 1, 8 and 15, carboplatin on all three days. Phase 2: four AC cycles every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 1, 3, 5 and 8; after surgery, every 42 days for five doses. See the whole-treatment map for monitoring and sequencing.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	Phase 1 · weeks 1–12	Before / during AC	Surgery / postoperative	After treatment
Cancer medicines NCCP 00861	Confirm diagnosis and suitability	Four 21-day cycles: weekly paclitaxel; carboplatin AUC 1.5 weekly. Pembrolizumab C1 and C3 Day 1.	Four AC cycles every 14 days. G-CSF each cycle. Pembrolizumab C5 and C8 Day 1.	Surgery → pathology; pembrolizumab 400 mg every 42 days ×5 if appropriate	Review late effects and follow-up
Surgery EARLY · NEO	Discuss operation options	After systemic treatment	Plan after recovery	Lumpectomy or mastectomy as agreed	Surgical follow-up
Radiotherapy EARLY	Assess indication	Not a routine concurrent chemotherapy phase	Plan with surgery / radiation teams	After surgery if indicated; may overlap pembrolizumab if agreed	Late-effect review
Blood tests NCCP 00861	FBC, renal and liver; verify GFR	Weekly before planned doses	Before each 14-day AC cycle	Before each pembrolizumab cycle	Symptoms or abnormalities determine follow-up
ECG / echo NCCP 00861 · ESC	ECG and cardiac risk review; echo if indicated	Symptoms prompt earlier assessment	Review before first doxorubicin; further echo by risk / symptoms	Review any cardiac symptoms or previous abnormalities	Risk-based cardiac follow-up

Care / check	Before starting	Phase 1 · weeks 1–12	Before / during AC	Surgery / postoperative	After treatment
Imaging / markers EARLY · NEO	Breast/nodal imaging; staging as indicated	Clinical response review; image for concern	Assess response and plan surgery	Pathology review; agree next steps	Breast surveillance; no routine CT/ markers when asymptomatic
Thyroid / hormones NCCP 00861	Thyroid baseline; hepatitis screening per source	Thyroid every 3–6 weeks	Continue immune monitoring	Continue while on pembrolizumab	Delayed immune symptoms still require review

What each check is for

1

Blood tests

FBC, renal and liver tests weekly during the carboplatin/paclitaxel phase, then before every AC cycle. Continue the required blood tests before each postoperative pembrolizumab cycle.

Check safety before the planned dose.

Ask / act: Agree the booking owner, result reviewer and plan if delayed.

2

ECG / cardiac review

Baseline ECG and cardiac history/risk assessment. NCCP lists baseline echo/multigated acquisition (heart pumping) scan (MUGA) for age over 65. Recheck the cardiac assessment before starting doxorubicin; echo/MUGA during treatment is driven by symptoms, risk or the agreed cardio-oncology plan. There is no automatic three-monthly echo schedule for every person on this pathway.

Check risk before and during anthracycline exposure.

Ask / act: Agree the booking owner, result reviewer and plan if delayed.

3

Cancer imaging

Record baseline breast/nodal findings and stage with imaging appropriate to the presentation. Assess clinical response during neoadjuvant treatment and use the appropriate breast imaging before surgery or sooner for concern. There is no universal three-monthly CT schedule for this early-breast pathway. After local treatment, arrange the agreed breast surveillance; routine CT/positron emission tomography (PET) and tumour markers are not asymptomatic recurrence screening.

Assess response and plan local treatment.

Ask / act: Agree the booking owner, result reviewer and plan if delayed.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan

Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks). Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. G-CSF support is required for every dose-dense AC cycle; the team prescribes its product and timing separately. After surgery: pembrolizumab 400 mg every 42 days for five doses, beginning six weeks after cycle 8 Day 1, subject to recovery, surgery and the oncology plan. Progression preventing definitive surgery or unacceptable pembrolizumab toxicity changes this plan; postoperative treatment is not automatic.

STEP 1

Before the first visit

Blood tests, ECG and cardiac review, medicines review, fertility discussion where relevant and the consent conversation come first. Include the prescribed immune baseline checks.

STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

Trained staff check and access the line, give the prescribed premedicines and infusions, and monitor for reactions. Tell the nurse immediately about pain, burning, rash or breathlessness.

STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

Visits include assessment, pharmacy preparation, premedicines, infusions and observation. The first visit and a change in phase can take longer. Ask about transport and any escort needed after sedation or line insertion.

Tests and checks through treatment

Ask who books each test, who reviews the result, and when you should hear the plan.



Blood tests

FBC, renal and liver tests weekly during the carboplatin/paclitaxel phase, then before every AC cycle. Continue the required blood tests before each postoperative pembrolizumab cycle.



ECG and heart ultrasound

Baseline ECG and cardiac history/risk assessment. NCCP lists baseline echo/MUGA for age over 65. Recheck the cardiac assessment before starting doxorubicin; echo/MUGA during treatment is driven by symptoms, risk or the agreed cardio-oncology plan. There is no automatic three-monthly echo schedule for every person on this pathway.

Thyroid and hormone checks

Baseline thyroid tests; thyroid monitoring every 3–6 weeks during pembrolizumab. Cortisol is checked when clinically indicated; tell the team about recent steroid use when results are interpreted. Immune side effects may appear after treatment ends.



Scans and tumour markers

Record baseline breast/nodal findings and stage with imaging appropriate to the presentation. Assess clinical response during neoadjuvant treatment and use the appropriate breast imaging before surgery or sooner for concern. There is no universal three-monthly CT schedule for this early-breast pathway. After local treatment, arrange the agreed breast surveillance; routine CT/PET and tumour markers are not asymptomatic recurrence screening.

Surgery, radiotherapy and follow-up

Other treatments depend on the complete diagnosis and pathology.

How the treatments fit

Chemotherapy comes before surgery. The surgeon discusses lumpectomy or mastectomy as alternatives according to your situation. Radiotherapy, if indicated, follows surgery and recovery; timing is agreed by both teams. It is not automatically given at the same time as these chemotherapy phases. Radiation may overlap postoperative pembrolizumab if both teams agree; this is not an automatic instruction.

Hormone tablets and bone scans

This is a triple-negative pathway: endocrine therapy and endocrine-related dual-energy X-ray absorptiometry (bone density) (DEXA) are not automatically part of it. Bone-density testing may still be appropriate for other personal risk factors.

If symptoms or tests change

The team may pause, change or stop a medicine, reassess imaging or revise the surgical plan. Report concerns promptly.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Phase 1 · four 21-day cycles

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
Carboplatin + paclitaxel · Day 1	 **Day 1** intravenous (IV) treatment
Carboplatin + paclitaxel · Day 8	 **Day 8** IV treatment
Carboplatin + paclitaxel · Day 15	 **Day 15** IV treatment
Pembrolizumab · C1 and C3 Day 1 only	 **Day 1** Separate immune schedule
Blood counts / infection	 **Days 1–21** Fever or sudden illness: call
Nerves / nails	 **Days 1–21** Report before each weekly visit
Immune symptoms	 **Days 1–21** Can occur on any day or after treatment

The first phase: Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks).

Changing to AC: Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. G-CSF support is required for every dose-dense AC cycle; the team prescribes its product and timing separately.

Surgery and afterwards: After surgery: pembrolizumab 400 mg every 42 days for five doses, beginning six weeks after cycle 8 Day 1, subject to recovery, surgery and the oncology plan. Progression preventing definitive surgery or unacceptable pembrolizumab toxicity changes this plan; postoperative treatment is not automatic.

> **Why these bands stay broad:** Treatment markers show the planned days. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00861 v2 · official regimen ↗

> **There are no guaranteed "safe days".** Fever, shivers or suddenly feeling unwell require immediate contact. Serious symptoms override the timeline or diary.

Phase 2 · four 14-day cycles

Day 1 → Day 14: a general pattern

Treatment or check	Timing in this view
AC · Day 1	 **Day 1** IV treatment
Nausea / appetite	 **Days 1–5** Early or delayed effects
Blood counts / infection	 **Days 1–14** Fever or sudden illness: call
G-CSF after chemotherapy	 **Days 2–14** Exact days follow the prescription
Pembrolizumab · C5 and C8 Day 1 only	 **Day 1** Separate immune schedule

Treatment or check	Timing in this view
Immune symptoms	 **Days 1–14** Can occur on any day or after treatment

The first phase: Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks).

Changing to AC: Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. G-CSF support is required for every dose-dense AC cycle; the team prescribes its product and timing separately.

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> **Why these bands stay broad:** Treatment markers show the planned days. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00861 v2 · official regimen ↗

Postoperative · 42-day schedule

Day 1 → Day 42: a general pattern

Treatment or check	Timing in this view
Pembrolizumab · Day 1	 **Day 1** IV treatment
Immune symptoms	 **Days 1–42** Report during and after treatment

The first phase: Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks).

Changing to AC: Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. G-CSF support is required for every dose-dense AC cycle; the team prescribes its product and timing separately.

Surgery and afterwards: After surgery: pembrolizumab 400 mg every 42 days for five doses, beginning six weeks after cycle 8 Day 1, subject to recovery, surgery and the oncology plan. Progression preventing definitive surgery or unacceptable pembrolizumab toxicity changes this plan; postoperative treatment is not automatic.

> **Why these bands stay broad:** Treatment markers show the planned days. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00861 v2 · official regimen ↗

Between-treatment notes

Day 1–21: Follow the current phase

Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks). Your team books actual appointments and reviews results. Call about urgent symptoms on any day.

Support around the medicines

1 Separate anti-sickness plans

The platinum/taxane phase and AC need separate prescriptions. Carboplatin/paclitaxel is moderate emetic risk on each weekly treatment day. AC uses its own high-risk plan.

2 Growth-factor support

G-CSF is prescribed for every dose-dense AC cycle. Ask for the exact injection days and instructions.

3 Checks before the next dose

FBC, renal and liver tests weekly during the carboplatin/paclitaxel phase, then before every AC cycle. Continue the required blood tests before each postoperative pembrolizumab cycle. Keep the thyroid and other immune-monitoring appointments.

Questions worth asking

When should I call the oncology team?

Call immediately for fever, shivers, sudden illness, new breathing difficulty or concerning symptoms. Do not wait for a scheduled blood test or appointment.

Why is my chemotherapy changing halfway through?

The planned course uses a platinum/taxane phase followed by AC, which works in a different way. The change is part of the pathway and does not by itself mean the first medicines failed.

Will I need to enter dates here?

No. The maps explain the sequence. Your clinical team provides actual appointment dates.

When are ECGs, echoes and blood tests done?

Baseline ECG and cardiac history/risk assessment. NCCP lists baseline echo/MUGA for age over 65. Recheck the cardiac assessment before starting doxorubicin; echo/MUGA during treatment is driven by symptoms, risk or the agreed cardio-oncology plan. There is no automatic three-monthly echo schedule for every person on this pathway. FBC, renal and liver tests weekly during the carboplatin/paclitaxel phase, then before every AC cycle. Continue the required blood tests before each postoperative pembrolizumab cycle.

Does pembrolizumab follow every chemotherapy visit?

No. Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks). Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. G-CSF support is required for every dose-dense AC cycle; the team prescribes its product and timing separately. After surgery: pembrolizumab 400 mg every 42 days for five doses, beginning six weeks after cycle 8 Day 1, subject to recovery, surgery and the oncology plan. Progression preventing definitive surgery or unacceptable pembrolizumab toxicity changes this plan; postoperative treatment is not automatic.

> **Try explaining the plan back:** Which phase are you in, which medicines are active, and what checks are due before the next phase?

Pembrolizumab

- Immune inflammation of bowel, lungs, liver, kidneys or skin
- Hormone-gland problems, sometimes permanent
- Rare serious heart or neurological inflammation
- Infusion reaction
- Effects can first appear after treatment ends

Carboplatin

- Low blood counts, infection and bleeding
- Nausea, vomiting and fatigue
- Allergic reaction, sometimes after previous exposure
- Kidney or electrolyte changes

Paclitaxel

- Low blood counts and infection
- Numbness / tingling, sometimes lasting
- Hair loss, muscle or joint pain
- Mouth, bowel, skin and nail effects

Doxorubicin

- Low counts, infection and mouth ulcers
- Nausea, hair loss and fatigue
- Heart damage, sometimes delayed
- Severe tissue injury if the infusion leaks
- Rare later blood cancers

Cyclophosphamide

- Low counts and infection
- Nausea, vomiting and hair loss
- Bladder irritation / bleeding
- Fertility effects and menopause

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Phase 1 · weekly carboplatin / paclitaxel

Active medicines: Carboplatin + Paclitaxel + Pembrolizumab on its separate schedule

Emetogenic risk: Moderate · carboplatin AUC <4

Medicine	Dose / route	When
Ondansetron	8 mg orally twice daily	Day 1 NCCP alternative to 16 mg once daily; review QT / hepatic risks
Dexamethasone	8 mg orally	Day 1; reconcile paclitaxel premedication Do not add to the paclitaxel steroid course automatically
Dexamethasone	8 mg orally daily	Days 2–3 reference antiemetic row Reconcile the delayed period with paclitaxel steroids; one final individualised steroid prescription

Medicine	Dose / route	When
Metoclopramide	10 mg orally up to three times daily as required	Subsequent days: duration to be confirmed Review contraindications and renal function

Additional phase-specific support

- First paclitaxel dose: dexamethasone 8 mg intravenous (IV), chlorphenamine 10 mg IV and famotidine 20 mg IV 30 minutes before treatment.
- Day 8 and later: no routine dexamethasone unless the consultant specifies it for reaction/risk; chlorphenamine continues and any famotidine omission requires consultant/local-policy agreement.
- Reconcile antiemetic and taxane dexamethasone into one prescription; never add duplicate steroid courses.
- Four 21-day cycles: paclitaxel on Days 1, 8 and 15; carboplatin AUC 1.5 on Days 1, 8 and 15. Pembrolizumab 400 mg on Day 1 of cycles 1 and 3 only (every six weeks).
- This is a weekly combination visit, not paclitaxel alone.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

Phase 2 · dose-dense AC

Active medicines: Doxorubicin + Cyclophosphamide + Pembrolizumab on its separate schedule

Emetogenic risk: High · AC

Medicine	Dose / route	When
Aprepitant	125 mg orally	Day 1
Ondansetron	16 mg orally	Day 1 Individual dose / QT / hepatic review
Dexamethasone	12 mg orally	Day 1 No routine Days 2–4 dexamethasone in this NCCP AC antiemetic row
Olanzapine	5 mg orally	Days 1–4 Review sedation, falls, interactions and glucose
Aprepitant	80 mg orally daily	Days 2–3
Metoclopramide	10 mg orally up to three times daily	Subsequent days: duration and use to be confirmed Caution with olanzapine: additive extrapyramidal effects; adjust / avoid for individual contraindications

Additional phase-specific support

- Four further cycles (cycles 5–8): doxorubicin and cyclophosphamide on Day 1 every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 5 and 8 only; the six-week clock continues across the change in chemotherapy. granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) support is required for every dose-dense AC cycle; the team prescribes its product and timing separately.
- G-CSF every dose-dense AC cycle is required. Product, start day and duration must be prescribed using the local supportive-care policy; do not administer simultaneously with cytotoxic chemotherapy.
- Confirm cumulative doxorubicin exposure, cardiac assessment and venous-access plan.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

After surgery · pembrolizumab alone

Active medicines: Pembrolizumab

Emetogenic risk: Minimal

No routine prophylactic antiemetic order is populated for this phase. Agree an individual symptom-treatment plan where needed.

Additional phase-specific support

- After surgery: pembrolizumab 400 mg every 42 days for five doses, beginning six weeks after cycle 8 Day 1, subject to recovery, surgery and the oncology plan. Progression preventing definitive surgery or unacceptable pembrolizumab toxicity changes this plan; postoperative treatment is not automatic.
- No routine cytotoxic antiemetics, taxane premedicines or chemotherapy granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) are carried forward.
- Continue the immune-toxicity and required blood/hormone monitoring.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

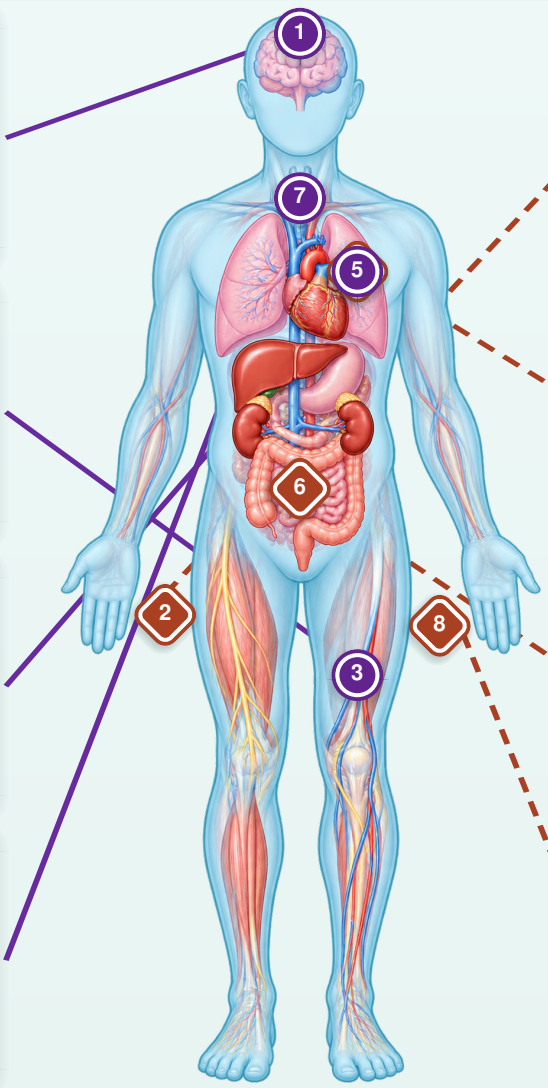
Pembrolizumab 400 mg + weekly carboplatin AUC 1.5 / paclitaxel → dose-dense AC

Phase 1: four 21-day cycles; paclitaxel on Days 1, 8 and 15, carboplatin on all three days. Phase 2: four AC cycles every 14 days. Pembrolizumab 400 mg on Day 1 of cycles 1, 3, 5 and 8; after surgery, every 42 days for five doses. See the whole-treatment map for monitoring and sequencing.

Source reconciled; independent oncology, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.



1 Hair / scalp

Paclitaxel

Hair loss is common; ask about support and scalp cooling where suitable.

2 Nerves / nails

Paclitaxel

Report numbness, tingling or painful nails before the next dose.

3 Blood / infection

Carboplatin and paclitaxel

Fever, shivers, unusual bleeding or sudden illness: call immediately.

4 Infusion reaction

Carboplatin or paclitaxel

Tell staff immediately about flushing, chest tightness or breathing difficulty.

5 Breathing

Pembrolizumab

New cough or breathlessness needs prompt assessment.

6 Bowel

Pembrolizumab

Report diarrhoea early, especially with pain, bleeding or poor intake.

7 Hormone glands / energy

Pembrolizumab

Report marked weakness, dizziness, headache or unusual thirst.

8 Skin / liver symptoms

Pembrolizumab

A severe rash, blistering or jaundice needs prompt advice.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

This printed map shows phase 1. See the next page for all medicines and the online phase-specific maps. Report persistent effects after a phase ends.

> Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00861 v2 · official regimen**
   https://healthservice.hse.ie/documents/6453/861_v2_Pembro_Carbo_AUC_1.5_Pacli_80_DDAC.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
4. **ASCO · neoadjuvant breast therapy and response assessment**
   https://pmc.ncbi.nlm.nih.gov/articles/PMC8274745/
5. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
6. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
7. **HSE · chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
8. **NCCP antiemetics v8, Table 1 pp4–5**
   https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf